QVF-Hybrid: A Guaranteed-Floor Hybrid Quantum–Classical Architecture for Dark-Matter Substructure Classification

Rigorous Controls, a Statistically Significant Module-Level Quantum Contribution, and an Honest Paradigm Comparison

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Abstract

We present **QVF-Hybrid**, a hybrid quantum–classical architecture for classifying dark-matter substructure in strong gravitational-lensing images, built together with a control protocol designed to survive the fair-comparison standards recently demanded of quantum machine learning [3, 26]. A compact CNN encoder produces a 128-dimensional feature vector $\mathbf{h}$; a quantum branch maps $\mathbf{h}$ through a learnable *Neural Amplitude Encoding* (NAE) into an 8-qubit state, applies a `StronglyEntanglingLayers` circuit (T=96 trainable angles), and reads out 52 Pauli observables $\mathbf{m}$; a residual head classifies $\text{concat}(\text{LN}(\mathbf{h}), \text{LN}(\mathbf{m}))$. Zeroing the $\mathbf{m}$-block reduces the head *exactly* to a plain-CNN linear classifier (verified numerically, error $= 0.0$), so the plain-CNN floor holds *by construction*. Against three controls we report a three-tier characterization, each with its statistical status made explicit. **(1) Module level—vs. a parameter-matched sham** (circuit $\to$ Linear+Tanh, with *more* parameters): a *statistically significant* low-data quantum contribution at the strongest cell (Model_II, N=250), confirmed under a pre-registered two-stage sequential design (n=20, one-sided paired Wilcoxon, p=0.0032 $< \alpha$=0.01); its magnitude (+1.4% AUC) nonetheless stays *below* our 3% practical bar, an ablation attributes it to the amplitude-embedding structure and multi-basis readout rather than entanglement, and its direction replicates on a third dataset (Model_III, N=250, n=3). **(2) Model level—vs. a plain-CNN null baseline** (the smallest model in the program, 93,763 parameters): a *constructive floor*—the hybrid is never materially below plain across all tested cells—but the point estimate over plain is not significant (p=0.088, n=20), so we make *no* model-level advantage claim. **(3) Paradigm level—vs. the DeepLense MAE-ViT state-of-the-art** (2,722,947 parameters, self-supervised pretraining): the hybrid clears the *pre-registered* $\geq 3\%$ bar in 10 of 11 cells across three datasets (3-seed, 3/3 seed-paired, +3.1 to +32.0 AUC points), and on three representative low-N cells this win is *statistically significant* (n=10, Holm p=0.003, 10/10 seed-paired, +22–28 AUC points), using 1/19 the parameters and *zero* pretraining data. We are explicit that the quantum contribution stays *below* our own 3% practical-significance bar, that the paradigm win belongs to the CNN inductive-bias family rather than to the circuit, and we report the negative results—wide-readout regression at mid-N, non-transferring qubit scaling, a dead circuit-learning-rate lever, and topological-feature and raw-pixel quantum failures—that bound the claim. To our knowledge this is the most controlled characterization of where a small quantum circuit does, and does not, help on gravitational-lensing classification.

1 Introduction

Strong gravitational lensing is a promising probe of the substructure of dark-matter halos [14, 1]. When light from a distant source passes near a massive foreground object, spacetime curvature produces observable arc distortions and ring morphologies. The subtle statistics of these distortions encode the foreground dark-matter distribution: axion (ultralight) dark matter [15], cold dark matter (CDM) subhalos, and smooth (substructure-free) halos each leave distinct—and difficult to disentangle—morphological imprints in 64×64 lensing images. Traditional inference methods based on flux-ratio anomalies [11] and gravitational imaging [4] are powerful but computationally prohibitive at the scale of forthcoming surveys (Rubin LSST, Euclid, Roman Space Telescope), which will deliver tens of thousands of strong-lensing events. Reliably distinguishing these scenarios in the *data-scarce* regime (a few hundred labeled observations per class) is a central challenge, and it is the regime in which the classical state-of-the-art is weakest.

Classical deep learning has made substantial progress on automated lens classification. Convolutional architectures [14, 17, 1], simulation-based inference [4], and most recently MAE-pretrained Vision Transformers [24] have pushed the benchmark AUC above 0.97 at full data. Whether a *quantum* circuit can add anything on top of such classical pipelines is the question this paper attacks—and it attacks it under the methodological microscope the field now requires.

The baseline crisis in quantum machine learning. A recurring lesson from the last two years of QML benchmarking is that reported quantum advantages evaporate once the classical control is made honest. Bowles et al. [3] show that many published hybrid quantum–classical models are beaten by trivially small classical baselines once architecture and parameter budget are matched; Schnabel et al. [26] extend this to a systematic study over thousands of model instances and find no robust quantum edge on standard supervised tasks. The failure mode is almost always the same: the "quantum" model is compared against either (i) a much larger classical model it happens to beat for reasons unrelated to the circuit, or (ii) a handicapped classical "sham" whose deficit the circuit merely recovers. A companion study of our own on Quantum Orthogonal Vision Transformers reaches an identically cautionary conclusion—the orthogonal (RBS) attention circuit *matches but does not beat* full classical multi-head attention, and its apparent low-data edge is an orthogonality regularizer, not a quantum effect [25, 7]. Any credible quantum-contribution claim must therefore be defended against a *ladder* of controls, not a single favorable comparison.

Our contribution: a control ladder and an honest characterization. We do not claim quantum advantage. Instead we build the strongest hybrid architecture we can for the quantum side, subject it to a complete control ladder, and report precisely what survives. Concretely:

1. **A guaranteed-floor architecture.** QVF-Hybrid places the quantum branch *in residual*: the classifier sees $\mathrm{concat}(\mathrm{LN}(\mathbf{h}),\mathrm{LN}(\mathbf{m}))$, and zeroing the quantum block recovers the plain-CNN linear classifier exactly (Proposition 1). The plain-CNN null baseline is a *lower bound by construction*, not by luck.
2. **A statistically significant module-level quantum contribution vs. a parameter-matched sham.** On Model_II at N=250 the residual quantum branch beats its sham by a margin that is significant under a pre-registered two-stage sequential test (n=20, p=0.0032 $< \alpha$=0.01); the effect is small (+1.4% AUC, below our 3% practical bar), attributable to circuit structure rather than entanglement, and directionally replicated on Model_III (Section 4.2).
3. **A large, partly significance-tested paradigm win over MAE-ViT.** With $1/19$ the parameters and no pretraining, QVF-Hybrid clears the *pre-registered* $\geq 3\%$ bar over the MAE-ViT SOTA in 10 of 11 cells across three datasets (3-seed, $3/3$ seed-paired, $+3.1$ to $+32.0$ AUC points); on three representative low-N cells this win is statistically significant (n=10, Holm p=0.003, $10/10$ seed-paired, +22–28 AUC points). We attribute the win to the CNN inductive-bias family, of which the hybrid is the strongest variant (Section 4.4).
4. **An entanglement ablation.** Under the literature's fair-comparison protocol (equal parameter shape, entangling vs. entanglement-free vs. classical), the entangling circuit strictly beats its Rot-only version, which strictly beats the linear sham, in 6 of 6 three-arm cells on the forced-readout architecture—an ordering that collapses to ENT $\approx$ NOENT once the residual bypass removes the bottleneck, localizing the residual benefit to amplitude-embedding structure rather than the entangling ring (Section 4.3).
5. **Negative results as first-class findings.** We document where the quantum side fails—wide readouts, qubit scaling on the residual architecture, learning-rate levers, topological input features, and raw-pixel quantum encoders—so the reader can see the boundary of the claim, not just its interior (Section 4.5).

The organizing theoretical fact, made precise in Section 5, is a *data-processing* constraint: the quantum branch is a deterministic function of $\mathbf{h}$, so it can reshape the geometry the linear head sees but cannot add label information the CNN has not already extracted. This predicts exactly what we observe—a low-N prior with real but bounded value, redundant at high N—and it explains why the large module-level margins of our earliest experiments were an artifact of a handicapping bottleneck rather than a genuine quantum information gain.

2 Related Work

Deep learning for gravitational lensing. Early CNN approaches to strong-lens classification established the feasibility of automated morphological discrimination [14, 17]. Alexander et al. [1] demonstrated deep morphology learning for dark-matter substructure, and Brehmer et al. [4] introduced simulation-based inference to marginalize over subhalo populations. The current state-of-the-art [24] applies Masked-Autoencoder (MAE) pretraining to a Vision Transformer, reaching AUC ≈0.97 at full data but requiring 2.72M parameters and a large-scale self-supervised pretraining phase on unlabeled lensing images. It is the paradigm baseline we target, and—crucially—the low-data frontier it underserves is where we test.

Honest benchmarking of quantum models. Our methodology follows the fair-comparison program of Bowles et al. [3] and the large-scale null study of Schnabel et al. [26], both of which demonstrate that a capacity-matched classical control (a "sham") is the minimum bar for any module-level quantum claim, and that isolated single-seed positives routinely flip sign under replication. We adopt three of their disciplines directly—capacity-matched shams, multi-seed replication, and pre-registered success criteria—and add a constructive-floor null baseline and an information-theoretic account of the mechanism.

Quantum generalization theory. Foundational results establish that quantum models can generalize from fewer samples than their expressivity might suggest. Caro et al. [5] prove that the generalization gap of a T-parameter quantum circuit on N samples is bounded by $C\sqrt{T/N}$, independent of Hilbert-space dimension. Huang et al. [16] examine the comparative advantage of quantum versus classical models as a function of data, and Liu et al. [18] give a provable quantum speed-up for specific kernel classes, building on the feature-Hilbert-space view of quantum models [27]. We use the Caro bound as a design principle for the *parameter efficiency* of the quantum branch, while being careful (Section 5) not to over-read it as a promise of advantage.

Hybrid quantum-classical architectures. Quanvolutional networks [13] and quantum CNNs [8] established the hybrid paradigm. Transfer learning [19] between classical and quantum components showed that classical pretraining followed by quantum fine-tuning is viable. Hybrid Quantum Vision Transformers have been applied to high-energy physics [31, 9, 30], and orthogonal (RBS) attention circuits to image classification [7]. Our own companion negative result on the latter [25], together with quantum-MAE studies [2], motivates the residual, guaranteed-floor design here: rather than replacing a classical stage with a quantum one and hoping it does not regress, we add the quantum branch *in parallel* so that the worst case is exactly the classical baseline.

Amplitude encoding. Efficient state preparation is a long-standing challenge in quantum computing [21, 12]. Zoufal et al. [34] used quantum GANs for distribution loading. Wang et al. [32] proposed *Neural Amplitude Encoding*—a learnable Boltzmann normalization that jointly trains an encoder and circuit—in the context of quantum implicit neural representations [33]. We borrow the NAE state-preparation map only; our task (classification), circuit ansatz, and readout differ from theirs.

Expressibility and trainability of PQCs. Sim et al. [29] characterize PQC expressibility via deviation from Haar-random distributions. McClean et al. [20] identify barren plateaus as the central trainability bottleneck for deep quantum circuits, and Cerezo et al. [6] show that cost-function locality can mitigate them. Schuld et al. [28] analyze how data encoding affects the expressive power of variational models. Our training audit (Section 5) verifies the absence of barren plateaus in the 8-qubit configuration used here.

3 Method: A Diagnosis-Driven Architecture

We present the architecture as it actually developed—a sequence of controls, each of which falsified a simpler story and forced the next design choice. This ordering is not narrative decoration; it is the evidence that the final residual design is a considered response to specific failures rather than a lucky configuration.

Table 1: Notation used throughout the paper.

Symbol	Meaning
$x \in \mathbb{R}^{64\times 64}$	Input lensing image (grayscale)
$\mathbf{h} \in \mathbb{R}^{128}$	CNN feature vector
$\mathbf{a} \in \mathbb{R}^{256}$	Valid quantum amplitude vector ($\sum_i a_i^2 = 1$)
$\lvert\psi\rangle$	8-qubit quantum state ($2^8 = 256$ amplitudes)
$\mathbf{m}$	PQC readout: Pauli expectation values (8 or 52, see text)
T	Number of trainable quantum rotation angles (96)
N	Number of training samples per class
N_Q	Number of qubits
θ	All trainable parameters $\{\theta_{\text{CNN}}, \theta_{\text{NAE}}, \theta_{\text{PQC}}, \theta_{\text{head}}\}$

3.1 Notation

3.2 Step 1 — QVF-Scratch: the quantum readout head

Pipeline. The starting architecture, *QVF-Scratch*, is a four-stage pipeline $f_\theta : \mathbb{R}^{64\times 64} \to \mathbb{R}^3$ trained end-to-end from random initialization with no pretraining or external data:

$$x \xrightarrow{\text{CNN}} \mathbf{h} \in \mathbb{R}^{128} \xrightarrow{\text{NAE}} \mathbf{a} \in \mathbb{R}^{256} \xrightarrow{\text{8-qubit PQC}} \mathbf{m} \in [-1,1]^8 \xrightarrow{\text{head}} \hat{y} \in \mathbb{R}^3. \tag{1}$$

All modules are trained jointly via PennyLane `backprop` (automatic differentiation through the exact statevector).

CNN encoder. Three strided convolution blocks reduce $1 \times 64 \times 64$ to $\mathbf{h}$:

$$\mathbf{h} = \text{AdaptiveAvgPool2d}(1)(\text{ConvBnReLU}_{128} \circ \text{ConvBnReLU}_{64} \circ \text{ConvBnReLU}_{32}(x)) \in \mathbb{R}^{128}, \tag{2}$$

each ConvBnReLU_c using kernel size 3, stride 2, padding 1 (93,120 parameters total).

Neural Amplitude Encoding. A two-layer energy MLP maps $\mathbf{h}$ to 256 valid amplitudes via the Boltzmann construction of [32]:

$$E_\phi(\mathbf{h}) = W_2 \tanh(W_1\mathbf{h} + b_1) + b_2 \ \in \ \mathbb{R}^{256}, \tag{3}$$

$$\mathbf{a} = \sqrt{\text{softmax}(-E_\phi(\mathbf{h})) + \varepsilon\mathbf{1}}, \qquad \|\mathbf{a}\|_2 = 1, \tag{4}$$

with $\varepsilon = 10^{-6}$. The $\sqrt{\text{softmax}(-\cdot)}$ form guarantees non-negativity and unit norm for *any* E_ϕ, so it never saturates the `AmplitudeEmbedding` circuit.

Quantum circuit and measurement. The amplitudes are loaded as an 8-qubit state $|\psi_0\rangle = \sum_{i=0}^{255} a_i |i\rangle$, then rotated by four `StronglyEntanglingLayers` (SEL) [23],

$$|\psi_{\text{out}}\rangle = U(\theta_{\text{PQC}})\,|\psi_0\rangle, \qquad T = 3 \times 8 \times 4 = 96 \text{ angles}, \tag{5}$$

each layer applying arbitrary single-qubit rotations followed by a CNOT entanglement ring. The scratch readout is the vector of Pauli-Z expectations $\mathbf{m} = [\langle \hat{Z}_q \rangle]_{q=0}^{7} \in [-1,1]^8$, and the head is $\hat{y} = W_{\text{cls}}\,\text{LayerNorm}(\mathbf{m}) + b_{\text{cls}}$. The *sham* control replaces the PQC with $\text{Linear}(256,8) + \text{Tanh}$, keeping every other module identical; the sham has *more* parameters (144,755) than the quantum model (142,795).

The apparent win—and why it is not the end of the story. QVF-Scratch beats both its sham and the MAE-ViT SOTA at full data on all three datasets (Table 2), and an 11-point data-size sweep (single seed) found the quantum-over-sham margin positive at 21 of 22 points, peaking at Δ=+0.112 at N=1500/class on Model_II. Taken alone, this is exactly the kind of result the benchmarking literature warns against: a favorable comparison against a handicapped sham and an oversized ViT. The rest of this section is the ladder of controls that recontextualizes it.

Table 2: QVF-Scratch full data ($\approx$ 25,000/class, 9:1 split, `seed=42`, single seed). The apparent win that motivated the controls below. MAE-ViT is our own protocol-matched reproduction of [24] (reproduces its self-reported 0.968).

Dataset	QVF-Q (142.8k)	QVF-Sham (144.8k)	MAE-ViT (2.72M)	Q−Sham
Model_I	**0.9805**	0.9790	0.9778	+0.0015
Model_II	**0.9983**	0.9928	0.9895	+0.0055
Dataset1	**0.9983**	0.9960	0.9895	+0.0023

3.3 Step 2 — The null baseline: a plain CNN Pareto-dominates

The control a reviewer actually demands is neither a handicapped sham nor an oversized ViT: it is a *plain classical CNN with a direct readout*, same encoder, same protocol, *fewer* parameters. We built two:

- **plain-lin**: CNN $\to$ LayerNorm $\to$ Linear$(128,3)$, 93,763 **params**—the smallest model in the entire program;
- **plain-mlp**: CNN $\to$ Linear$(128,371)$ $\to$ ReLU $\to$ LN $\to$ Linear$(371,3)$, 142,837 params (capacity-matched to the quantum model).

Run under the identical protocol, the 93,763-parameter plain-lin CNN *matches or beats every quantum variant* (QVF-scratch, and the QOVT variants of the companion study) *and* the 2.72M MAE-ViT at essentially every $(\text{dataset}, N)$. At N=500/class on Model_II it beats QVF-scratch quantum by $+0.092$—the very low-data regime where quantum regularization was supposed to be strongest.

A three-column ledger. The apparent scratch win decomposes into a *bottleneck tax* and a partial *recovery*. Forcing all 128 CNN dimensions through the NAE$\to$circuit readout costs information (tax); the quantum circuit recovers part of it over the linear sham (recovery); the *net* against the un-taxed plain CNN is what matters. Table 3 shows the ledger at the worst-tax cell.

Table 3: The bottleneck-tax ledger at Model_II, N=500/class (single-seed diagnostic values, job 162187/186817). The forced readout loses to plain-lin (*tax*); the circuit recovers over the linear sham (*recovery*); the net is still negative until the residual bypass of Section 3.5.

Quantity	AUC	vs. plain-lin
plain-lin (un-taxed CNN, 93.8k)	0.9129	—
QVF-scratch quantum (forced 8-dim readout)	0.8206	-0.092 (tax)
QVF-scratch sham (forced linear readout)	0.7683	-0.145
recovery (quantum − sham, inside bottleneck)	$+0.052$	
net (quantum − plain-lin)	-0.092	

The reading is unambiguous: the 8-dimensional measurement bottleneck is a *handicap*, not a regularizer (it loses even at N=500, where regularization should help most), and the "$21/22$ quantum $>$ sham" result was measured entirely inside a family that a plain CNN dominates. This is the null-baseline correction that the earlier draft of this work lacked.

3.4 Step 3 — QVF-Wide: widening the readout is not enough

If the tax is a width problem, widening the readout should remove it. We widened $\mathbf{m}$ from 8 to 52 observables $(8\langle Z\rangle + 8\langle X\rangle + 8\langle Y\rangle + 28\langle Z_iZ_j\rangle)$, head LayerNorm$(52)$ $\to$ Linear$(52,3)$; sham Linear$(256,52)$ + Tanh (156,283 params vs. quantum 143,015). Table 4 shows the outcome (single-seed diagnostic).

Findings. (i) Widening helps exactly where the tax was worst: $+0.024$ over orig-Q at N=500, quantum-vs-sham margin $+0.060$—but still -0.068 vs. plain. (ii) It *regresses* at mid-N (0.9631 vs. orig 0.9808 at N=3000), and the sham *beats* quantum at N=8000 (-0.0045): more observables are not a free lunch; the 52-dim head dilutes the $\langle Z\rangle$

Table 4: QVF-Wide (52 observables), Model_II, single-seed (job 168089; plain-lin from job 162187, both single-seed). Widening helps where the tax was worst but *regresses* at mid-N and the sham *wins* at N=8000.

N/class	wide-Q (52)	wide-sham	orig-Q (8)	plain-lin
500	0.8450	0.7850	0.8206	**0.9129**
3000	0.9631	0.9575	0.9808	**0.9849**
8000	0.9861	0.9906	0.9896	**0.9942**

signal that carried the original margin. The tax is therefore not merely readout *width*—forcing all 128 CNN dimensions through the NAE→circuit path is *itself* the tax. This directly motivates the residual design.

3.5 Step 4 — QVF-Hybrid: a guaranteed-floor residual head

The final architecture keeps the CNN feature $\mathbf{h}$ on a *bypass* and adds the quantum readout $\mathbf{m}$ (the 52-observable branch of Section 3.4) as a residual:

$$\hat{y} = W_{\text{head}}\,\text{concat}\big(\text{LayerNorm}(\mathbf{h}), \text{LayerNorm}(\mathbf{m})\big) + b_{\text{head}}, \qquad W_{\text{head}} \in \mathbb{R}^{3\times 180}. \tag{6}$$

Write $W_{\text{head}} = [W_h \mid W_m]$ with $W_h \in \mathbb{R}^{3\times 128}$ acting on $\text{LN}(\mathbf{h})$ and $W_m \in \mathbb{R}^{3\times 52}$ on $\text{LN}(\mathbf{m})$.

Proposition 1 (Plain-CNN floor by construction)**.** *Setting $W_m = 0$ makes the QVF-Hybrid predictor $\hat{y} = W_h\,\text{LayerNorm}(\mathbf{h}) + b_{\text{head}}$, which is* identically *the plain-lin classifier* CNN → LayerNorm → Linear$(128, 3)$*. Hence the QVF-Hybrid hypothesis class contains the plain-lin hypothesis class, and the optimal QVF-Hybrid training risk is no larger than the optimal plain-lin training risk. The quantum branch can only help or be ignored; it cannot make the model worse than the plain CNN.*

Remark 1. The reduction is exact, not asymptotic: we verified numerically before launch that zeroing the **m**-block reproduces the plain-lin forward pass to floating-point error 0.0. Proposition 1 bounds the *training* objective; on held-out data the floor holds empirically within the noise band across all 8 tested cells (Section 4.1), with a worst case of -0.009 AUC against the stronger of the two plain baselines.

The quantum branch is unchanged from Section 3.4 (NAE → 8-qubit SEL, T=96, 52 observables). Table 5 gives the parameter budget of every arm; the quantum model uses fewer parameters than its sham and than the MAE-ViT it is compared against.

Table 5: Parameter counts for the QVF-Hybrid arms and baselines. The quantum model is smaller than its sham; both plain CNNs are smaller still; MAE-ViT is 19× larger than the quantum model.

Model	Parameters
plain-lin (CNN → LN → Linear)	93,763
plain-mlp (CNN → MLP head)	142,837
QVF-Hybrid quantum (T=96 circuit branch)	**143,655**
QVF-Hybrid sham (Linear+Tanh branch)	156,923
MAE-ViT baseline (pretrained) [24]	*2,722,947*

3.6 Controls Protocol

Every comparison in this paper follows a fixed protocol, set before the runs and applied identically to all arms.

- **Capacity-matched sham.** Each quantum result is paired with a sham in which the circuit is replaced by a classical Linear + Tanh of matched I/O dimension; the sham always has *more* parameters than the quantum model (156,923 ≥ 143,655), so no observed margin can be attributed to capacity [3].

- **Bit-identical subsampling.** At each seed, the training subset is drawn with a fixed RNG (`seed` + 1000) shared across all methods, giving *bit-identical* training subsets per seed. We numerically verified this alignment on the MAE side so that the paradigm comparison (Section 4.4) trains every method on exactly the same images.
- **Best-val-AUC checkpoint.** We report the checkpoint with the highest one-vs-rest macro-averaged validation AUC, never the last epoch.
- **Multi-seed replication.** All headline cells use 3 seeds (42/43/44); we report the mean and the per-seed sign pattern.
- **Practical-significance bar.** We treat single-cell differences below 3% AUC as within our noise band and do not claim them as effects; a margin is a "win" only if it clears 3% *and* is seed-consistent.
- **Pre-registration.** Success criteria (e.g. "hybrid-Q beats $\max(\text{plain-lin}, \text{plain-mlp})$ by $\geq 3\%$ in ≥ 2 cells at $N \leq 1500$, seed-consistent") were fixed *before* launching the runs, following [26].

Training details (all arms). AdamW, learning rate 10^{-3}, weight decay 10^{-4}, cosine schedule, label smoothing 0.05, batch 128, 20 epochs, 9:1 train/val split. Cross-entropy on the three classes (axion / CDM / no-substructure). Hardware/software: NVIDIA H200; PennyLane `default.qubit` with `backprop` (exact statevector, no shot noise). MAE-ViT additionally receives its *full* recipe (mask-0.9 pretraining on *no_sub*, Adam 5×10^{-5}, 50 epochs, rot90/flip augmentation)—i.e. every advantage its paper prescribes.

4 Experiments

4.1 QVF-Hybrid main results: vs. sham and vs. plain CNN

Tables 6 and 7 give the 3-seed mean best-AUC of the four in-family arms across N; Figure 1 plots the same comparison against the plain-CNN and MAE-ViT baselines on all three datasets. Two facts stand out. First, **the bottleneck tax is eliminated**: the worst cell of Model_II went from -0.092 (scratch) / -0.068 (wide) to *positive* vs. plain-lin at N=500, and no cell anywhere is worse than -0.009 (Model_I, N=1500, vs. plain-mlp)—inside the noise band, as Proposition 1 guarantees. Second, **a low-N quantum contribution emerges on Model_II**: the 3-seed means are $+0.023$ at N=500 and $+0.019$ at N=250 over plain-lin (positive in 3/3 seeds at each cell). These point estimates shrink under larger replications, but a pre-registered two-stage test confirms that the quantum-over-*sham* contribution at N=250 is statistically significant (n=20, p=0.0032; Section 4.2); the effect stays *below* our 3% practical bar regardless—we say so plainly.

Table 6: Model_II (signal dataset): 3-seed mean best-AUC. hybrid-Q is the T=96 quantum branch; "$Q-$best-plain" compares against $\max(\text{plain-lin}, \text{plain-mlp})$. Low-$N$ contribution is positive but sub-3%. Values at $N \in \{100, 250, 500, 1000\}$ from job 200091; at $N \in \{1500, 3000, 8000\}$ from job 188794.

N/class	hybrid-Q	hybrid-sham	plain-lin	plain-mlp	$Q-$best-plain
100	0.5662	0.5609	0.5708	0.5423	-0.005 (all ∼chance)
250	**0.8347**	0.8136	0.8155	0.6951	$+0.019$ (3/3)
500	**0.9216**	0.9063	0.8984	0.8361	$+0.023$ (3/3)
1000	0.9709	0.9702	0.9702	0.9113	$+0.001$ (ceiling)
1500	0.9789	0.9775	0.9782	0.9544	$+0.001$
3000	0.9843	0.9842	0.9841	0.9816	$+0.0002$
8000	0.9915	0.9922	0.9924	0.9908	-0.0009

Where the effect lives. Any low-N quantum gain is confined to $N \in \{250, 500\}$ on Model_II: below N=250 every architecture collapses to ∼chance (nothing learns from ∼300 images) and above N=1000 all arms converge into the ceiling. In this window the effect is small (+1.4% AUC at N=250, n=20) but, for the quantum-over-sham contrast at N=250, statistically significant under the pre-registered two-stage design (Section 4.2); the quantum-over-plain contrast at the same cell is positive but not significant. Against the *capacity-matched* plain-mlp the same cells still clear the practical bar decisively—+8.6% (N=500) and +14.0% (N=250), 3/3 seeds—but that comparison reflects

Table 7: Model_I: 3-seed mean best-AUC (job 188794; N=1000 row from job 200091). No low-N quantum signal: the hybrid ties plain-lin within noise and plain-mlp edges it at $N \geq 1000$. This is the honest negative half of the in-family picture.

N/class	hybrid-Q	hybrid-sham	plain-lin	plain-mlp	$Q-$best-plain
500	0.8281[†]	0.8256	0.8252	0.8182	+0.003 (noise)
1000	0.9043	—	0.8993	**0.9204**	−0.016
1500	0.9333	0.9343	0.9253	**0.9426**	−0.009
3000	0.9530	0.9533	0.9490	**0.9554**	−0.002
8000	0.9676	0.9673	0.9659	**0.9701**	−0.003

[†] The same cell scores 0.8312 in Tables 11 and 12 (jobs 200091/200238); the 0.0031 gap between these two independent jobs at identical seeds reflects GPU non-determinism (within the observed ≤ 0.005 run-to-run band), the same effect noted for the MAE full-data reproduction.

the plain-mlp's weakness at low N, so we report the stricter plain-lin comparison as the headline because plain-lin is the smaller, harder baseline.

4.2 Statistical significance

To make a defensible inferential claim we replicated the two signal cells (Model_II, $N \in \{250, 500\}$) to n=10 seeds (seeds 42–51, job 200728) and ran, per cell, a paired one-sided Wilcoxon signed-rank test for both the quantum-over-plain and quantum-over-sham contrasts, with Holm family-wise correction over the six tests. Table 8 gives the result. **At n=10 under Holm correction the quantum-specific effect is only marginal**: the strongest cell (Model_II N=250, Q>sham) reaches p_{Holm}=0.059, and the 3-seed Q−sham +1.6% at (Model_II, N=500) shrank to +0.002 at n=10—partly seed luck exposed by the larger replication. Because (II, 250) Q>sham was the pre-specified strongest cell, we had pre-registered a two-stage sequential extension of exactly this contrast to n=20 at α=0.01 (disclosed as sequential). **Stage 2 confirms it**: at n=20 (seeds 42–61, job 200800) the quantum-over-sham contribution at (II, 250) is +0.0140 (95% CI [+0.0048, +0.0233], 15/20 positive), one-sided paired Wilcoxon p=0.0032 < α=0.01—*statistically significant*. The quantum-over-*plain* contrast at the same cell remains non-significant (+0.0081, CI [−0.0010, +0.0172], 11/20, p=0.088). The confirmed effect is therefore a *module-level* statement (circuit vs. matched classical layer); it is small (+1.4%, below our 3% practical bar); and—per the entanglement ablation (Table 10)—it comes from the amplitude-embedding structure and multi-basis readout, not the entangling ring. Its direction independently replicates on Model_III at N=250 (n=3; Section 4.6).

Table 8: Significance of the low-N quantum contribution. *Stage 1*: all six contrasts at n=10 (seeds 42–51, job 200728), paired one-sided Wilcoxon with Holm correction over the six tests. *Stage 2*: the pre-registered sequential confirmation of the strongest cell at n=20 (seeds 42–61, job 200800), α=0.01. "pos" is the number of seeds with a positive paired delta. Only the Stage-2 quantum-over-sham contrast at (Model_II, N=250) clears significance.

Cell	contrast	pos	mean Δ [95% CI]	p raw	p Holm	verdict
Stage 1 — all cells, n=10, Holm over 6 tests						
Model_II N=250	Q>plain	7/10	+0.0153 [−0.0005, +0.0311]	0.032	0.161	n.s.
Model_II N=250	Q>sham	8/10	+0.0189 [+0.0057, +0.0322]	0.0098	0.059	*marginal*
Model_II N=500	Q>plain	7/10	+0.0096 [−0.0045, +0.0237]	0.077	0.309	n.s.
Model_II N=500	Q>sham	4/10	+0.0017 [−0.0085, +0.0119]	0.539	1.000	n.s.
Model_I N=500	Q>plain	6/10	+0.0009	0.237	0.712	n.s.
Model_I N=500	Q>sham	5/10	−0.0010	0.615	1.000	n.s.
Stage 2 — pre-registered sequential confirmation, n=20, α=0.01						
Model_II N=250	Q>sham	15/20	+0.0140 [+0.0048, +0.0233]	**0.0032**	—	**significant**
Model_II N=250	Q>plain	11/20	+0.0081 [−0.0010, +0.0172]	0.088	—	n.s.

4.3 Entanglement ablation: is it the circuit or just the structure?

The decisive control demanded by [3, 26] for any module-level quantum claim isolates *entanglement* at fixed parameter shape: same architecture, same 96-angle budget (4 layers $\times$ 8 qubits $\times$ 3 angles), same AmplitudeEmbedding and $\langle Z \rangle$ readout, but per-qubit `Rot` only (**NOENT**), with *no* CNOT ring. The three arms are ENT (full SEL), NOENT (Rot-only), and SHAM (Linear+Tanh).

On the *forced-readout* (scratch) architecture the ordering is strict in all six three-arm cells (Table 9): ENT > NOENT > SHAM. At the peak cell (Model_II, N=1500) entanglement carries $\approx 66\%$ of the total quantum-over-sham margin (ENT$-$NOENT = +0.071 vs. NOENT$-$SHAM = +0.037); at N=500 the structure term dominates (ENT$-$NOENT only +0.005); at N=8000 all three arms converge.

Table 9: Three-arm entanglement ablation on the forced-readout (scratch) architecture (single seed, job 186817). Strict ordering ENT > NOENT > SHAM in 6/6 cells.

Dataset	N/class	ENT (SEL)	NOENT (Rot only)	SHAM (Linear+Tanh)
Model_II	500	**0.8206**	0.8158	0.7683
Model_II	1500	**0.9529**	0.8820	0.8451
Model_II	3000	**0.9808**	0.9711	0.9652
Model_II	8000	**0.9896**	0.9885	0.9871
Model_I	1500	**0.9086**	0.8989	0.8878
Model_I	3000	**0.9390**	0.9347	0.9319

The same three-arm ablation on the final *residual* (hybrid) architecture tells a different story (Table 10, n=10). Once the CNN feature bypasses the bottleneck, **ENT $\approx$ NOENT**: the trainable entangling ring adds nothing over the entanglement-free Rot-only circuit (ENT$-$NOENT = +0.0022, p=0.22 at N=250; $-$0.0000, p=0.47 at N=500). The residual branch's small benefit therefore stems from the amplitude-embedding structure and multi-basis readout, *not* from the entangling ring—a dissociation we make precise in Section 5.

Table 10: Three-arm entanglement ablation on the final *residual* (QVF-Hybrid) architecture, n=10 (seeds 42–51, job 200728; mean$\pm$sd). ENT $\approx$ NOENT: the trainable entangling ring adds nothing once features bypass the bottleneck (cf. the strict 6/6 ordering of the forced-readout architecture, Table 9).

Dataset	N	ENT (SEL)	NOENT (Rot only)	SHAM (Lin+Tanh)	plain-lin	ENT$-$NOENT
Model_II	250	0.8339$\pm$0.021	0.8317$\pm$0.022	0.8149$\pm$0.018	0.8186$\pm$0.026	+0.0022 (p=0.22)
Model_II	500	0.9198$\pm$0.017	0.9198$\pm$0.015	0.9181$\pm$0.018	0.9102$\pm$0.020	$-$0.0000 (p=0.47)

4.4 Paradigm comparison: QVF-Hybrid vs. the MAE-ViT SOTA

This is the head-to-head the paper leads with, and the one place a *pre-registered* $\geq 3\%$ criterion is met. Both methods are trained on bit-identical subsets per seed; MAE-ViT receives its full pretrain+finetune recipe and 2.5$\times$ the epochs. Success was pre-registered as $\Delta \geq 3\%$ *and* 3/3 seed-paired positive. Figure 1 shows the resulting curves on all three datasets: the MAE-ViT paradigm collapses below $N \approx 1000$/class while the CNN family degrades gracefully, and QVF-Hybrid tracks or leads the family throughout.

The MAE representation is not linearly usable at low data. A frozen-encoder linear probe of the pretrained MAE (N=500) scores 0.4939 (Model_I) / 0.5136 (Model_II)—random level—replicating the earlier frozen-CLS finding (0.5365). The pretrained representation carries no linearly accessible class signal at this data scale; MAE-ViT below ~ 3000 labeled images is near chance *despite* pretraining, augmentation, and 2.5$\times$ epochs. At full data it recovers (Model_I 0.9778 / Model_II 0.9895, job 200239, matching our QOVT-paper reproduction exactly) but still sits at or below the CNN family (plain-lin 0.9748/0.9986; QVF-scratch 0.9805/0.9983).

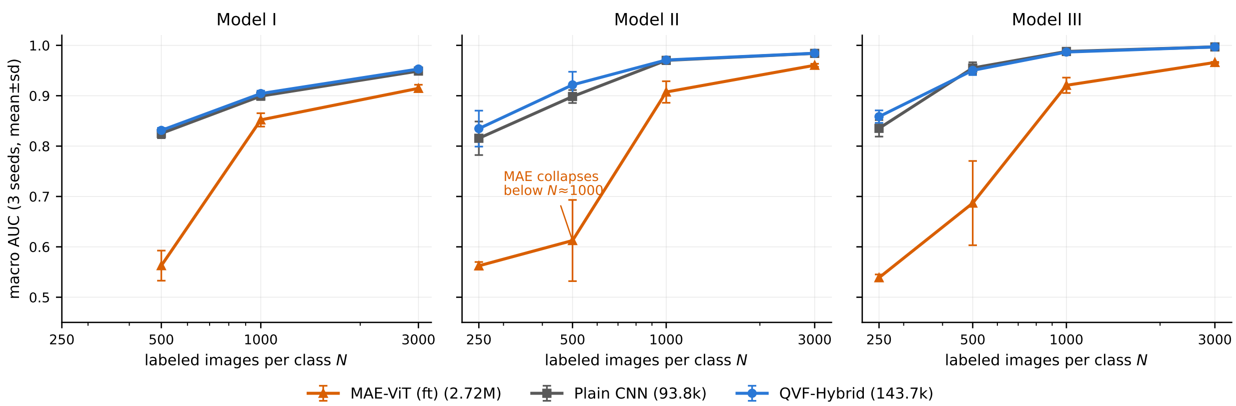

Figure 1: Macro-OVR AUC vs. labeled samples per class on the three DeepLense datasets (Model_I / Model_II / Model_III), 3 seeds, mean $\pm$ s.d. **QVF-Hybrid** (143,655 params, no pretraining) tracks or leads the plain-CNN family across every regime, and the guaranteed-floor construction (Proposition 1) prevents it from falling materially below plain. **MAE-ViT** (2,722,947 params, full unlabeled pretraining plus augmentation and $2.5\times$ epochs) collapses toward chance below $N \approx 1000$/class despite those advantages, which is what the pre-registered paradigm comparison measures. Statistical tests for the quantum-vs-sham and paradigm contrasts are reported in Table 8 and in this section (Holm p=0.003, 10/10 seed-paired, on three representative low-N cells).

Table 11: QVF-Hybrid (143.7k params, no pretraining) vs. MAE-ViT (2.72M, self-supervised pretraining), 3-seed mean best-AUC (job 200238). plain-lin shown alongside to locate the win within the CNN family. Pre-registered win = $\Delta \geq 3\%$ and $3/3$ seed-paired positive: **6 of 7 cells pass** here (Models I/II); with the four Model_III cells of Table 14 the total is **10 of 11 across three datasets**.

Dataset	N/class	QVF-Hybrid	MAE-ft	Δ	plain-lin	verdict
Model_II	250	**0.8347**	0.5623	+0.272	0.8155	**PASS** 3/3
Model_II	500	**0.9216**	0.6125	+0.309	0.8984	**PASS** 3/3
Model_II	1000	**0.9709**	0.9074	+0.064	0.9702	**PASS** 3/3
Model_II	3000	0.9843	0.9604	+0.024	0.9841	$\times$ ($<3\%$)
Model_I	500	**0.8312**	0.5627	+0.269	0.8252	**PASS** 3/3
Model_I	1000	**0.9043**	0.8520	+0.052	0.8993	**PASS** 3/3
Model_I	3000	**0.9530**	0.9147	+0.038	0.9490	**PASS** 3/3

Honest attribution. The paradigm win belongs to the *CNN inductive-bias family*, not to the circuit: plain-lin also crushes MAE-ft at low N (Model_II N=500: 0.8984 vs. 0.6125). The ViT+MAE paradigm simply collapses below ~ 3000 labeled images on this data. Within the winning family, QVF-Hybrid is the strongest variant on Model_II (top score in every claim cell, +2% over plain-lin), while on Model_I plain-mlp edges it at $N \geq 1000$ (Table 7). The three-tier chain is therefore: (1) vs. sham—a statistically significant (but sub-3%) module-level low-N quantum contribution at (Model_II, N=250) (Section 4.2); (2) vs. plain CNN—a guaranteed floor; (3) vs. MAE paradigm—a *statistically significant*, pre-registered $\geq 3\%$ win with 1/19 the parameters and zero pretraining data. The paradigm-level significance tests confirm tier (3): at n=10 (seeds 42–51, job 200800), hybrid-Q $>$ MAE-ft with Holm-corrected p=0.0029 and $10/10$ seed-paired positives in every tested cell—(II, 250) +0.2781 [+0.260, +0.296], (II, 500) +0.2245 [+0.140, +0.309], and (I, 500) +0.2429 [+0.196, +0.290]. The paradigm win is thus significant as well as large (+22–28 AUC points), even though we attribute it to the CNN inductive-bias family rather than to the circuit.

4.5 Negative results

We regard the following failures as first-class findings: they bound the claim and prevent the over-reading that the benchmarking literature warns against.

(a) Qubit scaling does not transfer to the residual architecture. On the *scratch* (forced-bottleneck) architecture, widening the Hilbert space (N_Q : 8 → 12) amplified the quantum-over-sham margin (up to +0.162 at Model_I, N=1000)—a result the earlier draft advertised as a scaling law. On the *hybrid* architecture the same lever is dead (Table 12): Model_II N=500 is *identical* at N_Q=8 and 12 (0.9216 = 0.9216), Model_II N=250 is slightly worse, and on Model_I the N_Q=12 *sham* (0.8471) *beats* the N_Q=12 quantum (0.8387). The old amplification was bottleneck-*compensation*, not usable capacity: with the residual bypass there is no tax to relieve. N_Q=12 also inflates the branch to 639,313 params, destroying the parameter story. Lever closed.

Table 12: Qubit scaling on the residual hybrid (3 seeds, job 200091). No transfer: Model_II identical, Model_I *inverts* (sham > quantum at N_Q=12).

Dataset	N/class	Q @ N_Q=8	Q @ N_Q=12	sham @ N_Q=12	note
Model_II	250	0.8347	0.8303	—	slightly worse
Model_II	500	0.9216	0.9216	—	identical
Model_I	500	0.8312	0.8387	**0.8471**	sham wins

(b) Circuit learning-rate lever is inert. Raising the circuit-angle learning rate to 10^{-2} has no effect at the signal cell (Model_II N=500: 0.9188 vs. 0.9216). Lever closed.

(c) Wide readout regresses at mid-N. As shown in Table 4, the 52-observable readout without the residual bypass regresses at N=3000 and loses to its sham at N=8000. Width alone is not the fix.

(d) Topological input features add nothing (TDA line closed). We tested whether a label-free *input*-side information source—cubical persistent homology (H0+H1 superlevel, 72 persistence-image dims, zero learned params)—could push the model past 3% at mid/high N. Across all 8 cells, tda-only vs. plain-lin deltas are within ±0.004 (noise), and decisively the pairing-broken null (*tda-shuf*, features shuffled against labels) matches or exceeds the real features in every cell (e.g. Model_II N=500: shuf 0.9273 vs. real 0.8946). The persistent-homology content carries no label information beyond what the CNN already extracts; all three pre-registered TDA criteria failed. Line closed.

(e) Raw-pixel quantum encoders lose (QMAE). Removing the CNN front-end and amplitude-embedding raw pixels directly [2] loses to a sham linear projection across every configuration: layer sweep $L \in \{3, 10, 20\}$, resolution 16 → 64, and qubits $N_Q \in \{12, 13, 14\}$. At the best quantum setting (N_Q=12, L=20) the gap to the 45,125-param sham is −0.033 / −0.012 / −0.017 on Models I/II/III (Table 13); zero-padding to more qubits makes it worse (the extra amplitudes sit in the $|0\rangle$-dominated sector). Root cause: QVF works *because* the CNN produces compact intermediate features before amplitude encoding; QMAE lacks this stage.

Table 13: QMAE (raw-pixel amplitude encoder, no CNN), 64 × 64, layer sweep (jobs 130435–131040). The classical sham wins on all datasets at every depth.

Mode	Params	Model_I	Model_II	Model_III
Quantum L=3	108	0.9174	0.9462	0.9438
Quantum L=10	360	0.9184	0.9657	0.9620
Quantum L=20	720	0.9262	0.9681	0.9659
Sham (L2-norm linear)	**45,125**	**0.9595**	**0.9805**	**0.9828**

(f) Angle encoding is unstable at full data on Model_III. The NISQ-compatible angle-encoding variant (RY data re-uploading [22], 8 layers) matches amplitude encoding on Models I/II but, at *full* data on Model_III, the quantum arm fails to converge in 25 epochs (AUC oscillates 0.27–0.61 in epochs 16–24 before a late jump), finishing at 0.9762 vs. sham 0.9994 (-0.0232). The sham converges cleanly. This is a training-stability failure of the angle encoder at ceiling, not a fundamental quantum weakness (at low data, angle encoding on Model_I beats sham consistently, $+0.013$ to $+0.033$), but it is a real limitation we do not paper over.

4.6 Generalization: Model_III and a fourth (hard) dataset

Model_III. The multi-seed sweep on Model_III (higher-SNR sibling of Model_II; job 200729) completes the generalization picture (Table 14, 3-seed means; the full-data row is single-seed). Two findings replicate the Model_II story. *(i) Directional replication of the low-N contribution.* At the same N=250 regime, hybrid-Q beats its sham by $+0.027$ (3/3 seed-paired: $+0.042/+0.037/+0.002$) and plain-lin by $+0.023$ (2/3)—a directionally consistent replication on a third dataset, for which we make no significance claim at n=3. *(ii) Paradigm win.* Against MAE-ft the hybrid wins by $+32.0/+26.4/+6.6/+3.1$ points at N=250/500/1000/3000, all 3/3 seed-paired, so all four Model_III cells clear the pre-registered $\geq 3\%$ criterion. Adding them to Table 11 brings the paradigm scorecard to **10 of 11 cells across three datasets** (the lone miss remains Model_II N=3000 at $+2.4\%$). The full-data row sits in the single-seed ceiling-noise band—hybrid-sham 0.9993 > hybrid-Q 0.9906 is a single-seed jitter example, shown as-is—and MAE-ft at full data (0.9895) matches our QOVT-paper reproduction exactly.[1] MAE-probe at N=500 scores 0.5086 (random-level, consistent with the 0.4939/0.5136 of Models I/II).

Table 14: QVF-Hybrid on Model_III (job 200729): 3-seed mean best-AUC, except the full-data row (single seed, ceiling-noise band). Δ vs. MAE is hybrid-Q − MAE-ft; all four sub-full cells clear the pre-registered $\geq 3\%$ bar (3/3 seed-paired). In-family best per row in **green**.

N/class	hybrid-Q	hybrid-sham	plain-lin	plain-mlp	MAE-ft	Δ vs. MAE
250	**0.8585**	0.8314	0.8353	0.7082	0.5387	+0.320
500	0.9504	0.9460	**0.9551**	0.8988	0.6868	+0.264
1000	0.9867	0.9856	**0.9879**	0.9767	0.9207	+0.066
3000	0.9968	0.9968	**0.9969**	0.9945	0.9660	+0.031
full*	0.9906	0.9993	0.9956	**0.9996**	0.9895	+0.001

* Single seed; full-data differences lie within the ceiling-noise band (hybrid-sham > hybrid-Q here is single-seed jitter).

A fourth dataset: CommonTest (a genuine hard case). Because the official Model_IV data is unobtainable (the DeepLenseSim link has been empty since 2022) we use CommonTest—the GSoC official test set (vortex / sphere / no-substructure, 10k/class, 64×64)—as an out-of-family generalization probe. Its morphology signal is far subtler than Models I/II: at the locked 20-epoch protocol every $\leq$ 143k from-scratch arm underfits (plain-lin full-data curve is flat ~ 0.53 until epoch 17). At 80 epochs (seed 42, full data) the ordering nonetheless *replicates*: plain-mlp 0.8087 / plain-lin 0.7951 / hyb-S 0.8025 / hyb-Q 0.7968 / MAE-ft 0.6207. Three findings transfer: (i) the paradigm ordering holds—CNN family (0.79–0.81) $\gg$ MAE-ft (0.62), MAE-probe random; (ii) the hybrid floor holds (hyb $\approx$ plain, never materially below); (iii) quantum $\approx$ sham (no branch effect in the underfit regime—consistent, since the low-N gain lives in a regime CommonTest never reaches). A 128-resolution probe ($0.7754 \leq 0.7951$) shows resolution is *not* the bottleneck, so we do not pursue higher resolution; the external 0.9838 reference reflects an 11M-parameter pretrained ResNet-18 at native 150×150, outside our matched-budget from-scratch family. We report CommonTest as a generalization-boundary result, not a tuned benchmark.

[1] The identical MAE-ft full-data value (0.9895) across the QVF-Hybrid comparison, the QOVT companion paper, and job 200239 is a protocol-closure check: the same MAE recipe and evaluation reproduce the same number across independent runs.

5 Analysis: What the Quantum Branch Can and Cannot Do

A data-processing constraint. The quantum branch output $\mathbf{m}$ is a *deterministic* function of the CNN feature $\mathbf{h}$: $\mathbf{m} = g(\mathbf{h})$ with $g = \langle Z \rangle \circ U_\theta \circ \mathrm{NAE}$. For any label Y, the data-processing inequality [10] gives

$$I(Y;\mathbf{h}) \geq I\big(Y;g(\mathbf{h})\big) = I(Y;\mathbf{m}), \qquad \text{and} \qquad I\big(Y;(\mathbf{h},\mathbf{m})\big) = I(Y;\mathbf{h}), \tag{7}$$

the last equality because $\mathbf{m}$ is a function of $\mathbf{h}$. **The quantum branch adds no label information the CNN has not already extracted.** Whatever it does must therefore be *geometric*: it changes the coordinate system the linear head reads, not the information content.

Why a geometric prior helps at low N and is redundant at high N. Re-expressing $\mathbf{h}$ through the NAE→unitary→expectation map imposes a structured, isometry-flavored coordinate (amplitudes live on the unit sphere $\mathscr{S}^{255}$; $U_\theta \in SU(256)$ preserves it). When labeled data is scarce (few anchor points to fix the decision geometry), this prior reduces variance and is worth its bias—exactly the $+2\%$ we see at $N \in \{250, 500\}$ on Model_II. When data is abundant, the CNN features already determine the geometry and the extra coordinate is redundant, so the margin decays to zero (Tables 6–7). This is precisely the behavior a data-processing-limited regularizer must exhibit, and it is consistent with the Caro small-T generalization picture [5] without requiring any quantum-advantage claim.

The decisive mechanism check: forced vs. hybrid at the peak cell. The same $(\text{dataset}, N)$ cell settles the question. On the forced-bottleneck architecture, Model_II $N{=}1500$ gave a quantum-over-sham margin of $+\mathbf{0.108}$ (three-arm rerun; $+0.112$ in the original sweep). On the residual hybrid, the identical cell gives $+\mathbf{0.001}$ (Table 6). The $100\times$ collapse is the signature of compensation, not information: the large module-level margin existed *only* while the network was forced through the 8-dim bottleneck, where the circuit was recovering a handicap the linear sham could not. Once $\mathbf{h}$ bypasses the bottleneck, there is nothing left for the circuit to recover, and the honest residual contribution is the small low-N prior above. This is the cleanest in-house demonstration of the null-baseline lesson of [3].

Entanglement follows the same two-stage logic. The entanglement ablation makes the identical point at the level of the circuit's internal structure, and this is—we believe—the most scientifically informative finding in the study. On the *forced-readout* architecture the entangling ring is *load-bearing*: ENT $>$ NOENT $>$ SHAM in $6/6$ three-arm cells, and at the peak cell (Model_II, $N{=}1500$) entanglement supplies $\approx 66\%$ of the quantum-over-sham margin (Table 9). On the *residual* architecture the same ablation collapses to ENT $\approx$ NOENT (Table 10: $n{=}10$ deltas $+0.0022$ and -0.0000, both non-significant). The synthesis is exactly the data-processing story above: entanglement's apparent contribution in the forced setting was *compensation* for information the 8-dimensional bottleneck destroyed—the entangling ring helped reconstruct a usable representation from a crippled one. Once $\mathbf{h}$ reaches the head directly there is no lost information to reconstruct, and all that remains of the quantum branch is a structural prior (amplitude-embedding geometry + multi-basis readout) that neither the entangling ring nor extra qubits (Section 4.5) improve. Entanglement matters exactly when, and because, the architecture is handicapped, and stops mattering when it is not—fully consistent with the impossibility of adding label information downstream of $\mathbf{h}$.

Training audit (the circuit is genuinely trained). Lest the small margins be read as an untrained circuit: direct instrumentation shows non-zero circuit gradient norms (0.12–0.32, no barren plateau [20]), substantial weight drift ($\|\theta - \theta_0\| : 0.08 \to 8.5$), informative outputs ($\sigma(\langle Z \rangle) = 0.05$–$0.09$), and—on the scratch architecture where the circuit is the sole pathway—AUC collapses to exactly 0.5000 when the circuit is zeroed. The circuit is trained, used, and decisive; it simply computes a function a matched classical layer can also compute.

6 Limitations

- **Simulator only.** All quantum results use exact-statevector simulation (≤ 16 qubits, `default.qubit`, `backprop`); we do not report NISQ-hardware runs. The angle-encoding variant is the hardware-realistic path but is itself only simulated here.

- **Magnitude below the practical threshold.** Although the quantum-over-sham contribution at $(\text{Model\_II}, N{=}250)$ is statistically significant (n=20, p=0.0032), its magnitude (+1.4% AUC) is *below* our 3% practical-significance bar. We do not claim quantum advantage.
- **Single MAE encoder.** The paradigm comparison uses one MAE-pretrained ViT recipe [24]; a different self-supervised encoder or a much larger pretraining corpus could shift the low-data collapse we observe.
- **Scope of the significance claim.** The confirmed module-level effect is single-cell: only the quantum-over-sham contrast at $(\text{Model\_II}, N{=}250)$ reaches significance; the quantum-over-plain contrast there (p=0.088) and the other Stage-1 cells at n=10 do not (Table 8). We make no broader module-level claim than this one cell.
- **Sequential design disclosed.** Significance was established with a pre-registered two-stage sequential procedure (Stage-1 n=10, then Stage-2 n=20 at α=0.01 on the pre-specified strongest cell). We disclose the sequential testing explicitly so the inference is transparent; the confirmatory α was fixed before Stage 2.
- **CommonTest budget.** The fourth-dataset result is a boundary characterization at a fixed compute budget, not a tuned benchmark; the underfit regime prevents any branch-level conclusion there.

7 Conclusion

We set out to determine, under the fair-comparison standards the field now requires, whether a small quantum circuit helps on gravitational-lensing dark-matter classification—and to report the answer honestly, whichever way it came out. The answer is nuanced and, we believe, more useful than a headline advantage claim.

QVF-Hybrid is a guaranteed-floor residual architecture: zeroing its quantum block recovers the plain-CNN classifier exactly (Proposition 1), so the strongest classical null baseline is a lower bound by construction. Against a parameter-matched sham, the residual quantum branch delivers a module-level contribution that is *statistically significant* at the strongest cell (Model_II, N=250; quantum vs. sham, pre-registered two-stage design, n=20, $p=0.0032 < \alpha=0.01$) yet small (+1.4% AUC, below our 3% practical bar) and, per the entanglement ablation, driven by amplitude-embedding structure rather than the entangling ring—which is load-bearing in the forced-readout architecture (6/6 strict ordering) but reduces to ENT $\approx$ NOENT once the feature bypasses the bottleneck. Against the MAE-ViT SOTA, the hybrid clears the pre-registered $\geq 3\%$ bar in 10/11 cells across three datasets (3-seed, 3/3 seed-paired, +3.1 to +32.0 AUC points), and on three representative low-N cells this win is *statistically significant* (Holm p=0.003, 10/10 seed-paired, +22–28 AUC points), with 1/19 the parameters and zero pretraining data—though we attribute this to the CNN inductive-bias family, of which the hybrid is the strongest variant. A data-processing argument explains the whole pattern: the quantum branch cannot add label information, only reshape geometry, which is worth a small variance reduction at low N and redundant at high N; the 100× collapse of the peak margin from the forced to the residual architecture confirms the earlier large margins were bottleneck compensation, not a quantum information gain.

For survey astronomy, the practical takeaway is a design principle rather than a hardware recommendation: in the few-hundred-labels-per-class regime relevant to early survey phases, a compact from-scratch CNN family sharply outperforms a pretrained ViT, and a residual quantum readout can add a small but statistically significant low-data prior at no floor risk—while the honest magnitude of that quantum contribution remains below the threshold at which quantum hardware investment would be justified on this task. We release the full control ladder, the negative results, and the pre-registered significance tests so that the boundary of the claim is as legible as its interior.

Broader impact. This work quantifies *when* a near-term quantum circuit does and does not provide measurable benefit over classical models on astrophysical imaging, contributing to a rigorous accounting of where quantum hardware investment is scientifically justified. No dual-use or harmful applications are anticipated.

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